

Internal Audit Report: NDMO Compliance (2024)

Company A - Internal Audit Department
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0.1	2024-11-05	Saeed Al-Yami	Initial draft - fieldwork findings
0.2	2024-11-15	Saeed Al-Yami	Added management responses & action plans
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1. Executive Summary

1.1 Audit Objective

The Internal Audit Department conducted a comprehensive audit of Company A's compliance with the National Data Management Office (NDMO) regulations for the 2024 calendar year. The audit was commissioned by the Board Audit Committee as part of the annual Internal Audit Plan and fulfills Company A's regulatory obligation to conduct independent assurance of NDMO compliance.

Primary Audit Objectives:

1. Assess Compliance: Evaluate Company A's compliance with all 191 NDMO controls across 15 domains
2. Identify Gaps: Identify gaps, deficiencies, or non-conformities in the compliance program
3. Test Effectiveness: Test the operational effectiveness of key controls and processes
4. Provide Assurance: Provide independent assurance to the Board and Executive Leadership on the maturity and effectiveness of the NDMO compliance program
5. Recommend Improvements: Identify opportunities for continuous improvement and best practice adoption

Audit Period: The audit covered compliance activities from January 1, 2024 through September 30, 2024, with fieldwork conducted from September through November 2024.

Regulatory Context: NDMO regulations establish mandatory standards for data management across Saudi Arabian public sector organizations. Compliance is assessed across 15 domains covering data governance, quality, security, privacy, and lifecycle management. Non-compliance can result in regulatory sanctions, financial penalties, and reputational harm.

1.2 Overall Conclusion

Overall Audit Rating: HIGH LEVEL OF MATURITY (92% Compliance)

Company A has achieved a High Level of Maturity in NDMO compliance, demonstrating strong governance, comprehensive policies and procedures, effective technology enablement, and robust operational controls. The organization has made significant progress since the 2023 audit (87% compliance), reflecting sustained investment in data management capabilities.

Key Strengths:

- Comprehensive Data Governance Framework: Well-defined roles, responsibilities, and decision rights with active Data Governance Committee and executive sponsorship from the Chief Data Officer (CDO)
- Mature Data Quality Program: Systematic data quality assessment, issue tracking, and remediation across critical data elements
- Strong Privacy & Security Controls: Robust PDPL compliance program with effective privacy impact assessments, consent management, and data subject rights processes
- Technology Enablement: Effective deployment of Informatica Data Quality, Informatica EDC (Enterprise Data Catalog), Collibra for business glossary, and RSA Archer for GRC
- Stakeholder Engagement: High levels of engagement from data stewards, business units, and IT teams with 96% training completion rate

Areas for Improvement:

- Legacy System Coverage: Metadata coverage for legacy systems (e.g., "Archive System X") is incomplete, though risk is low as these systems are read-only and sunset-planned
- Data Quality Rule Maturity: Some newer Critical Data Elements (CDEs) in Procurement domain lack defined data quality rules, creating risk for reporting errors
- Backup Testing Documentation: While backup procedures are robust and tested regularly, documentation of test results could be improved for audit trail purposes

1.3 Key Findings Summary

The audit assessed 191 NDMO controls across 15 domains through a combination of document review, stakeholder interviews, and sample testing. Findings were classified using the following categories:

Finding Category	Definition	Count	% of Total
Compliant	Control is designed effectively and operating as intended	175	92.1%
Observation	Minor gap or inefficiency that does not represent a risk	10	5.2%
Non-Conformity	Control gap that represents material risk and requires remediation	6	2.6%

Detailed Finding Breakdown:

10 Observations (Low Risk):

1. Metadata completeness for legacy "Archive System X" (Domain 13: Metadata Management)

2. Business glossary term approval workflow could be streamlined (Domain 2: Data Catalog)
3. Data lineage documentation for 3 Tier 3 systems incomplete (Domain 2: Data Catalog)
4. Data steward performance metrics not formalized in position descriptions (Domain 1: Data Governance)
5. Data retention schedule review frequency could be increased from annual to semi-annual (Domain 14: Data Retention)
6. Open Data Portal API documentation could be enhanced with more examples (Domain 15: Open Data)
7. Data classification labels not consistently applied in PowerPoint templates (Domain 7: Data Classification)
8. Data risk assessment refresh cycle could be reduced from annual to quarterly for critical systems (Domain 10: Risk Management)
9. Data sharing agreement template lacks specific NDMO control references (Domain 6: Personal Data Protection)
10. Master data golden record conflict resolution process could be better documented (Domain 4: Master & Reference Data)

5 Non-Conformities (Medium Risk):

1. Data Steward Training Deficiency: 3 newly appointed data stewards in Q3 2024 have not yet completed mandatory data steward bootcamp training (Domain 8: Training & Awareness)
2. Data Quality Rules Gap: 3 new Critical Data Elements (CDEs) in Procurement domain lack defined data quality rules, creating risk for reporting errors (Domain 3: Data Quality)
3. Backup Testing Documentation: Quarterly backup restoration tests are performed but results not systematically documented in GRC system (Domain 11: Data Security)
4. Privacy Impact Assessment Delay: 1 new system (Patient Portal 2.0) launched in Q3 2024 without completed Privacy Impact Assessment (PIA) - PIA conducted post-launch (Domain 6: Personal Data Protection)
5. Data Classification Audit Trail: Data classification changes not consistently logged in audit trail for 2 reclassified datasets (Domain 7: Data Classification)

No Critical Findings: The audit did not identify any critical findings representing severe or immediate risk to the organization.

1.4 Management Response

Management has accepted all audit findings and developed comprehensive action plans to address each observation and non-conformity. Key management commitments:

Non-Conformity Remediation (all closed or in progress):

- NC #1 (Data Steward Training): All 3 data stewards completed bootcamp training by November 15, 2024. Status: CLOSED
- NC #2 (DQ Rules Gap): Procurement data steward defined and implemented DQ rules for 3 CDEs by December 15, 2024. Status: CLOSED
- NC #3 (Backup Testing Documentation): IT Ops implemented automated documentation of backup test results in RSA Archer GRC system. Target: January 31, 2025. Status: IN PROGRESS
- NC #4 (PIA Delay): Privacy Officer completed retroactive PIA for Patient Portal 2.0 (November 2024). Process improvement: PIA gate added to project lifecycle. Status: CLOSED
- NC #5 (Classification Audit Trail): Data Governance team implemented audit logging for all classification changes in Collibra. Status: CLOSED

Observation Remediation (all in progress):

- All 10 observations have assigned owners, target dates (Q1-Q2 2025), and are tracked in the GRC system
- Management commits to closing all observations by June 30, 2025

Audit Committee Oversight:

- Audit Committee will receive quarterly updates on remediation progress
- Internal Audit will conduct follow-up review in Q2 2025 to validate closure of non-conformities

2. Audit Scope & Methodology

2.1 Scope of Audit

In-Scope:

- All 15 NDMO domains and 191 controls
- All systems, processes, and business units subject to NDMO regulations
- All data management activities from January 1, 2024 through September 30, 2024

Out-of-Scope:

- IT general controls (covered by separate IT audit)
- Financial statement audit (covered by external auditors)
- Vendor/third-party compliance (covered by vendor management audit)

NDMO Domains Assessed:

Domain #	Domain Name	Control Count	Priority
1	Data Governance	18 controls	HIGH
2	Data Catalog	15 controls	HIGH
3	Data Quality	16 controls	HIGH
4	Master & Reference Data	12 controls	MEDIUM
5	Data Architecture	10 controls	MEDIUM
6	Personal Data Protection	20 controls	HIGH
7	Data Classification	14 controls	HIGH
8	Training & Awareness	8 controls	MEDIUM
9	Audit & Compliance	11 controls	HIGH
10	Risk Management	13 controls	HIGH
11	Data Security	17 controls	HIGH
12	Supporting Documents	6 controls	LOW
13	Metadata Management	12 controls	MEDIUM
14	Data Retention	10 controls	MEDIUM
15	Open Data	9 controls	LOW
Total	15 domains	191 controls	-

2.2 Audit Objectives

The audit was designed to achieve the following specific objectives:

Objective 1: Design Effectiveness

Assess whether Company A's policies, procedures, and controls are appropriately designed to achieve NDMO compliance requirements.

Objective 2: Operating Effectiveness

Test whether controls are operating effectively in practice, not just documented on paper.

Objective 3: Maturity Assessment

Evaluate the maturity of Company A's data management capabilities against NDMO standards and industry benchmarks.

Objective 4: Gap Identification

Identify gaps, weaknesses, or non-conformities that create compliance risk.

Objective 5: Best Practice Recognition

Recognize areas of excellence and best practices that can be shared across the organization.

Objective 6: Improvement Recommendations

Provide actionable recommendations for continuous improvement.

2.3 Audit Approach

The audit employed a risk-based approach aligned with ISO 19011:2018 Guidelines for Auditing Management Systems and the Institute of Internal Auditors' International Professional Practices Framework (IPPF).

Audit Phases:

Phase 1: Planning (September 2024)

- Reviewed NDMO regulations and Company A compliance documentation
- Conducted risk assessment to prioritize audit areas
- Developed audit program and testing procedures
- Identified key stakeholders and scheduled interviews

Phase 2: Fieldwork (October-November 2024)

- Conducted 28 stakeholder interviews (see Appendix C)
- Reviewed 156 documents including policies, procedures, reports, and meeting minutes (see Appendix D)
- Performed sample testing of 50 randomly selected artifacts across high-priority controls
- Observed key processes (Data Governance Committee meeting, Data Quality Council, Privacy Impact Assessment review)
- Tested system configurations (Informatica EDC, Collibra, RSA Archer)

Phase 3: Reporting (November 2024)

- Analyzed findings and drafted audit report
- Obtained management responses and action plans
- Reviewed draft with Director of Internal Audit
- Presented to Audit Committee for approval

Audit Evidence:

All audit findings are supported by documented evidence including:

- Interviews with process owners
- Screenshots of system configurations
- Sample documents and records
- Process walkthroughs
- Independent testing results

2.4 Sampling Methodology

Due to the large volume of data management activities, the audit employed statistical sampling for testing control effectiveness:

Sample Selection:

- Population: All NDMO control activities during audit period (Jan-Sept 2024)
- Sample Size: 50 items selected randomly from high-priority controls
- Confidence Level: 95% confidence, 5% margin of error
- Sampling Method: Simple random sampling using random number generator

Sample Coverage:

Domain	Population Items	Sample Items	% Coverage
Data Governance	48 DGC meetings/decisions	5	10.4%
Data Catalog	4,520 catalog assets	8	0.2%
Data Quality	1,247 DQ issues	10	0.8%
Personal Data Protection	100 DSARs, 95 PIAs	12	8.6%
Data Security	12 security incidents	5	41.7%
Other Domains	Various	10	Varies

Detailed sample selection criteria are documented in Appendix B.

2.5 Audit Team

Name	Role	Certification	Responsibilities
Saeed Al-Yami	Lead Auditor	CIA, CISA	Overall audit leadership, Domain 1-6 assessment
Nora Al-Shehri	Senior Auditor	CISA, CRISC	Domain 7-11 assessment, IT security testing
Fahad Al-Manea	Auditor	CDMP	Domain 12-15 assessment, data quality testing
Yasmin Al-Qahtani	Director of Internal Audit	CIA, CFE	Quality review, stakeholder management

Independence: The audit team is independent of the data management function and has no operational responsibilities for NDMO compliance. The audit was conducted in accordance with the Institute of Internal Auditors' Code of Ethics and Standards.

3. Background & Context

3.1 NDMO Regulatory Framework

The National Data Management Office (NDMO) is a Saudi regulatory body established in 2022 to oversee national data management standards across public sector organizations. NDMO regulations are mandatory for all government entities, including healthcare organizations like Company A.

NDMO Regulatory Structure:

15 Domains:

1. Data Governance
2. Data Catalog
3. Data Quality
4. Master & Reference Data
5. Data Architecture
6. Personal Data Protection
7. Data Classification
8. Training & Awareness
9. Audit & Compliance
10. Risk Management
11. Data Security
12. Supporting Documents
13. Metadata Management
14. Data Retention
15. Open Data

191 Controls:

Each domain contains specific controls addressing policy, process, technology, and governance requirements. Controls are categorized by type:

- Strategic Controls: Governance, policy, roles & responsibilities
- Operational Controls: Processes, procedures, data lifecycle management
- Technology Controls: Systems, tools, automation, monitoring
- Assurance Controls: Audit, monitoring, reporting, continuous improvement

Compliance Assessment Levels:

NDMO defines 5 maturity levels for compliance assessment:

Level	Description	Characteristics
Level 1: Initial	Ad-hoc, reactive	No formal processes, manual, inconsistent
Level 2: Developing	Some processes defined	Basic documentation, limited automation
Level 3: Defined	Processes documented & standardized	Comprehensive policies, some automation

Level 4: Managed	Processes measured & controlled	Metrics-driven, mature automation
Level 5: Optimized	Continuous improvement culture	Proactive, industry-leading, innovative

Company A Target: Achieve Level 4 (Managed) maturity across all domains by end of 2025.

3.2 Company A's Compliance Program

Company A established a formal NDMO compliance program in 2022 following the enactment of NDMO regulations. The program is led by the Chief Data Officer (CDO) and supported by the Data Management Office (DMO).

Governance Structure:

Board of Directors

- Ultimate accountability for NDMO compliance
- Reviews annual NDMO compliance report
- Approves data management strategy and major policy changes

Audit Committee (Board Committee)

- Oversees Internal Audit's independent assurance of NDMO compliance
- Reviews audit findings and management responses
- Monitors remediation of non-conformities

Executive Leadership (CDO)

- Ahmed Al-Rashid, Chief Data Officer - appointed January 2023
- Accountable for NDMO compliance program
- Chairs Data Governance Committee
- Reports quarterly to Board on compliance status

Data Governance Committee (DGC)

- Members: CDO (Chair), CFO, CMO, CIO, Legal Counsel, Privacy Officer
- Meets monthly
- Approves policies, procedures, and major data decisions
- Reviews NDMO compliance metrics
- Escalates material issues to Board

Data Management Office (DMO)

- 12 FTE dedicated to data management (Data Architects, Data Quality Analysts, Metadata Managers, Data Catalog Administrators)
- Led by Fatima Al-Zahrani, DMO Manager
- Day-to-day execution of data management activities
- Coordinates with data stewards across business units

Data Stewards (12 positions)

- One data steward per major business unit/domain
- Part-time (20% FTE), embedded in business units
- Responsible for data quality, metadata, data access governance within their domain

- Report to business unit leaders with dotted-line to DMO

Compliance Framework:

Company A's NDMO Compliance Framework (v2.0, approved March 2024) defines:

- Roles & responsibilities for NDMO compliance
- Policy hierarchy (strategy, policies, procedures, guidelines)
- Control design and operating effectiveness testing approach
- Compliance reporting requirements
- Continuous improvement process

3.3 Prior Audit History

2023 NDMO Compliance Audit (AUD-2023-008):

- Audit Date: Q4 2023
- Overall Rating: Developing Maturity (87% compliance)
- Findings: 18 observations, 7 non-conformities
- Status: All non-conformities closed by Q2 2024

Key Improvements Since 2023:

1. Hired dedicated Chief Data Officer (Ahmed Al-Rashid) in January 2023
2. Expanded DMO from 5 FTE to 12 FTE
3. Implemented Informatica EDC data catalog (February 2024)
4. Launched Open Data Portal (February 2024)
5. Achieved 96% training completion rate (up from 78% in 2023)
6. Reduced data quality issues from 1,847 (2023) to 1,247 (2024) - 32% improvement

External Regulatory Reviews:

NDMO conducted an on-site regulatory review in June 2024:

- NDMO Assessment: Level 3 (Defined) maturity - compliant with 89% of controls
- Action Items: 8 recommendations, all closed by September 2024
- Next NDMO Review: June 2026 (biennial cycle)

4. Compliance Assessment by Domain

4.1 Domain 1: Data Governance

Domain Overview:

Data Governance establishes the organizational framework, roles, responsibilities, policies, and decision rights for data management. It provides the foundation for all other NDMO domains.

NDMO Requirements (18 controls):

- DG-01: Data Governance Strategy approved by Board
- DG-02: Data Governance Committee established with executive representation

- DG-03: Chief Data Officer (CDO) appointed with clear accountability
- DG-04: Data Management Office (DMO) established with adequate resources
- DG-05: Data Steward network established across business units
- DG-06 through DG-18: Policies, procedures, performance metrics, continuous improvement

Compliance Assessment:

Control	Status	Notes
DG-01: DG Strategy	■ Compliant	DM_PDP_Strategy_2025_v2.0 approved by Board in Dec
DG-02: DGC Established	■ Compliant	DGC meets monthly with 95% attendance rate, robust
DG-03: CDO Appointed	■ Compliant	Ahmed Al-Rashid appointed January 2023, reports to
DG-04: DMO Established	■ Compliant	12 FTE in DMO (Data Architects, DQ Analysts, Metad
DG-05: Data Stewards	■■ Observation	12 data stewards appointed across business units.
DG-06: Data Governance Policy	■ Compliant	Data Governance Policy v3.1 approved January 2024,
DG-07: Decision Rights	■ Compliant	RACI matrix clearly defines decision rights for da
DG-08: Escalation Process	■ Compliant	Documented escalation path from data stewards → DM
DG-09: Stakeholder Engagement	■ Compliant	Quarterly data steward forums, annual data managem
DG-10: Performance Metrics	■ Compliant	Comprehensive DG metrics dashboard (16 KPIs) revie
DG-11: Budget & Resources	■ Compliant	SAR 3.5M annual budget for data management, fully
DG-12: Training Program	■ Compliant	Comprehensive training program with 96% completion
DG-13: Communication Plan	■ Compliant	Monthly DMO newsletter, quarterly DGC updates to s
DG-14: Policy Review Cycle	■ Compliant	All policies reviewed annually, review log maintai
DG-15: Continuous Improvement	■ Compliant	Quarterly DG maturity self-assessments, lessons le
DG-16: External Benchmarking	■ Compliant	Participated in NDMO peer benchmarking study (June
DG-17: Vendor Governance	■ Compliant	Data Processing Agreements (DPAs) in place for 95
DG-18: Change Management	■ Compliant	Data-related changes follow documented change mana

Overall Domain Rating: 94.4% Compliance (17/18 controls compliant, 1 observation)

Audit Testing Performed:

- Reviewed DGC meeting minutes for all 9 meetings (Jan-Sept 2024)
- Interviewed CDO (Ahmed Al-Rashid) and DMO Manager (Fatima Al-Zahrani)
- Tested 5 randomly selected data governance decisions for adherence to RACI matrix
- Reviewed Data Governance Policy v3.1 for completeness vs. NDMO requirements
- Verified data steward appointment letters and position descriptions

Key Strengths:

- Executive Sponsorship: Strong executive leadership from CDO with direct reporting to CEO
- Active Governance: DGC meets monthly with high attendance and robust action tracking
- Comprehensive Documentation: Policies, procedures, and guidelines are comprehensive and well-maintained

■ Stakeholder Engagement: High levels of engagement from data stewards and business units

Observation:

■■ Data Steward Performance Metrics: While data steward position descriptions define roles and responsibilities, they do not include formal performance metrics (e.g., data quality improvement targets, metadata completeness goals). This creates ambiguity for performance evaluation.

Management Response:

HR will update data steward position descriptions to include formal performance metrics by Q1 2025. DMO will develop standardized steward scorecard template.

4.2 Domain 2: Data Catalog

Domain Overview:

Data Catalog provides a centralized inventory of data assets, enabling users to discover, understand, and access data across the organization.

NDMO Requirements (15 controls):

- DC-01: Enterprise data catalog tool implemented
- DC-02: Data asset inventory comprehensive and current
- DC-03: Business glossary established
- DC-04: Data lineage documented
- DC-05 through DC-15: Search capabilities, access governance, metadata enrichment

Compliance Assessment:

Control	Status	Notes
DC-01: Catalog Tool	■ Compliant	Informatica EDC implemented February 2024, integra
DC-02: Asset Inventory	■ Compliant	4,520 data assets cataloged (databases, tables, fi
DC-03: Business Glossary	■■ Observation	3,890 business terms defined in Collibra. Observat
DC-04: Data Lineage	■■ Observation	Data lineage documented for Tier 1 and Tier 2 syst
DC-05: Search Functionality	■ Compliant	Robust search with 6,120 searches in Q4 2024, aver
DC-06: Access Governance	■ Compliant	Data access requests routed through catalog with a
DC-07: Data Ownership	■ Compliant	All assets assigned to data owners/stewards (100%
DC-08: Classification Tags	■ Compliant	All assets tagged with classification level (SECRE
DC-09: Usage Analytics	■ Compliant	Usage analytics tracked - most-accessed datasets,
DC-10: Data Profiling	■ Compliant	Informatica IDQ profiles data quality statistics,
DC-11: API Access	■ Compliant	Catalog API available for programmatic access (2.4
DC-12: User Training	■ Compliant	Catalog user training provided to 342 users, 4.2/5
DC-13: Metadata Standards	■ Compliant	Metadata standards defined (ISO 11179), consistent
DC-14: Catalog Refresh	■ Compliant	Automated daily metadata refresh from source syste
DC-15: Quality Assurance	■ Compliant	Quarterly catalog quality audits, 94.8% metadata q

Overall Domain Rating: 86.7% Compliance (13/15 controls compliant, 2 observations)

Audit Testing Performed:

- Logged into Informatica EDC and tested search functionality
- Reviewed sample of 8 cataloged assets for metadata completeness
- Interviewed Data Catalog Administrator (Nouf Al-Ghamdi)
- Reviewed business glossary approval workflow in Collibra
- Tested data lineage for 3 randomly selected critical data elements

Key Strengths:

- Comprehensive Coverage: 4,520 assets cataloged across 58 data sources
- High User Adoption: 342 registered users with 85 daily active users
- Technology Maturity: Informatica EDC is enterprise-grade with strong metadata management capabilities

Observations:

- Business Glossary Approval Workflow: Average 12-day approval time for new business terms is longer than industry benchmark (5-7 days). Workflow involves 4 sequential approvals which could be parallelized.
- Data Lineage for Tier 3 Systems: Data lineage is incomplete for 3 low-priority Tier 3 systems (Archive System X, Legacy HR System, Old Finance System). Risk is low as these systems are read-only and sunset-planned.

Management Responses:

- DMO will streamline business glossary approval workflow to 3 parallel approvals, targeting 7-day average by Q1 2025
- DMO will complete data lineage for 3 Tier 3 systems by Q2 2025 (included in system sunset project plan)

4.3 Domain 3: Data Quality

Domain Overview:

Data Quality ensures data is accurate, complete, consistent, timely, and fit for purpose across the organization.

NDMO Requirements (16 controls):

- DQ-01: Data Quality Policy approved
- DQ-02: Critical Data Elements (CDEs) identified
- DQ-03: Data quality rules defined for CDEs
- DQ-04: Data quality monitoring and reporting
- DQ-05 through DQ-16: Issue management, root cause analysis, remediation tracking

Compliance Assessment:

Control	Status	Notes
DQ-01: DQ Policy	■ Compliant	Data Quality Policy v2.3 approved March 2024
DQ-02: CDEs Identified	■ Compliant	147 Critical Data Elements identified across 8 bus
DQ-03: DQ Rules Defined	■ Non-Conformity	NC #2: 3 new CDEs in Procurement domain lack defin
DQ-04: DQ Monitoring	■ Compliant	Informatica IDQ runs daily DQ scans, dashboards av

DQ-05: DQ Scorecard	■ Compliant	Comprehensive DQ scorecard with 12 dimensions (acc
DQ-06: Issue Logging	■ Compliant	All DQ issues logged in centralized register (1,24
DQ-07: Issue Prioritization	■ Compliant	Issues prioritized by business impact (Critical, H
DQ-08: SLA Management	■ Compliant	DQ issue resolution SLAs defined and tracked (94%
DQ-09: Root Cause Analysis	■ Compliant	Root cause analysis performed for all High/Critica
DQ-10: Remediation Tracking	■ Compliant	Remediation actions tracked in GRC system with aut
DQ-11: DQ Council	■ Compliant	Data Quality Council meets monthly, chaired by DMO
DQ-12: Data Profiling	■ Compliant	Systematic data profiling for new data sources bef
DQ-13: DQ Reporting	■ Compliant	Monthly DQ reports to DGC, quarterly reports to Bo
DQ-14: Continuous Improvement	■ Compliant	DQ improvement initiatives tracked, 32% reduction
DQ-15: Training	■ Compliant	DQ training provided to data stewards and IT teams
DQ-16: Technology Tools	■ Compliant	Informatica IDQ fully deployed for DQ profiling, m

Overall Domain Rating: 93.8% Compliance (15/16 controls compliant, 1 non-conformity)

Audit Testing Performed:

- Reviewed Data Quality Policy v2.3 for alignment with NDMO requirements
- Tested 10 randomly selected DQ issues for proper logging, prioritization, and resolution tracking
- Interviewed Data Quality Lead (Mohammed Al-Ghamdi)
- Reviewed DQ Council meeting minutes (9 meetings in 2024)
- Logged into Informatica IDQ and reviewed DQ rules configuration
- Verified DQ scorecard calculations for 3 critical data elements

Key Strengths:

- Mature DQ Program: Comprehensive DQ framework with clearly defined processes, tools, and governance
- Technology Enablement: Informatica IDQ provides robust automation for DQ profiling and monitoring
- Continuous Improvement: 32% YoY reduction in DQ issues demonstrates sustained progress

Non-Conformity:

■ DQ Rules Gap for New CDEs: 3 new Critical Data Elements were added to the Procurement domain in Q3 2024:

1. "Vendor Tax ID Number" (added August 2024)
2. "Contract Expiration Date" (added August 2024)
3. "Supplier Performance Rating" (added September 2024)

Per NDMO Control DQ-03, all CDEs must have defined data quality rules (completeness, accuracy, validity, consistency, timeliness). These 3 CDEs are missing DQ rule definitions, creating risk for:

- Reporting errors in procurement dashboards
- Compliance violations (vendor tax reporting)
- Contract management failures (missed renewals)

Management Response:

Procurement data steward (assigned in August 2024) was not aware of requirement to define DQ rules immediately upon CDE designation. Data steward has now defined and implemented DQ rules for all 3 CDEs:

- Vendor Tax ID Number: Must be 10-digit numeric, validated against government registry
- Contract Expiration Date: Must be future date, alert 90 days before expiration
- Supplier Performance Rating: Must be 1-5 scale, validated against approved rating criteria

Status: CLOSED (as of December 15, 2024)

Process improvement: DMO updated CDE designation procedure to include mandatory DQ rule definition checklist. Training provided to all data stewards on DQ rule requirements (November 2024).

(Continuing with remaining 11 domains in similar detail...)

5. Detailed Findings

5.1 Observation #1: Legacy System Metadata Gaps

Finding Classification: Observation (Low Risk)

NDMO Domain: Domain 13 - Metadata Management

NDMO Control: MM-04 (Technical Metadata Completeness)

Condition:

Technical metadata (table schemas, column definitions, data types, relationships) is incomplete for legacy "Archive System X," a read-only archival database containing historical patient records from 1995-2010.

Criteria:

NDMO Control MM-04 requires "comprehensive technical metadata for all data assets cataloged in the enterprise data catalog, including legacy and archival systems."

Cause:

Archive System X was not originally built with data dictionary documentation. The system was developed in-house in 1995 using proprietary technology. Original developers have retired and documentation was not maintained. Manual reverse-engineering of the database schema is resource-intensive.

Effect:

- Users cannot easily discover what data is available in Archive System X
- Data lineage from Archive System X to downstream reports is incomplete
- Risk of missing responsive records for FOI requests or litigation holds

Risk Assessment:

Risk Level: LOW

Mitigating Factors:

- Archive System X is read-only (no data updates, inserts, or deletes)
- System is sunset-planned for decommissioning in Q3 2025 (data will be migrated to modern data warehouse)
- Usage is minimal (5 users, <10 queries/month)
- Critical data elements from Archive System X have been manually documented

Management Response:

Action Plan:

DMO will manually catalog critical tables and columns in Archive System X by Q1 2025:

- Owner: DMO Data Catalog Administrator (Nouf Al-Ghamdi)
- Target Date: March 31, 2025
- Resources: 40 hours of manual cataloging work
- Approach: Focus on 15 most-accessed tables first, then expand to full database

Alternatively, accelerate Archive System X decommissioning from Q3 2025 to Q2 2025, eliminating need for metadata cataloging.

Audit Conclusion:

Finding classified as Observation (Low Risk) due to mitigating factors and planned remediation. Recommend management prioritize system decommissioning over manual cataloging to avoid expending resources on sunset system.

5.2 Observation #2: DQ Rules for New CDEs

(Reclassified from Observation to Non-Conformity based on management discussion)

Finding Classification: Non-Conformity #2 (Medium Risk)

NDMO Domain: Domain 3 - Data Quality

NDMO Control: DQ-03 (Data Quality Rules for CDEs)

(See Section 4.3 for full details - remediation completed December 15, 2024)

(Continuing with remaining findings...)

Appendices

Appendix A: Audit Testing Summary

(Detailed testing procedures and results for all 50 sample items)

Appendix E: NDMO 191 Controls Checklist

(Complete checklist showing compliance status for all 191 controls)

END OF REPORT

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Distribution:

- Board Audit Committee
- Chief Executive Officer
- Chief Data Officer
- Executive Leadership Team
- NDMO (External Regulator)

Follow-Up Audit: Q2 2025 (validation of non-conformity closures)